

Infrastructure Economics and Management

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Low Altitude Economics

Week 5

Learning Objectives

By the end of this class, you should be able to:

- **Compare** different vertiport archetypes and identify the main site-selection trade-offs.
- **Estimate** how throughput, reliability, access time, and operating cost affect infrastructure viability.
- **Calculate** breakeven activity and evaluate a staged ramp-up under uncertain demand.
- **Assess** alternative methods for allocating scarce slots while protecting priority services.
- **Evaluate** hub, distributed, and hybrid network designs for Hong Kong and the wider GBA.

The Infrastructure Bottleneck

Imagine planning a new LAE service in Hong Kong.

- Aircraft may be available, but the service also needs a safe operating site, power, communications, passenger or cargo access, maintenance support, and sufficient throughput.
- A shortage of suitable infrastructure can constrain the entire service network.

Managerial question: Which resource is most likely to become the binding constraint: aircraft, landing capacity, charging, airspace access, or customer access?

- **Quick poll (2 min):** What is the most important infrastructure bottleneck for your Week 4 use case?

Session 1

Vertiport Types, Site Selection & Throughput Economics

Concept Primer: Three Vertiport Archetypes

For managerial analysis, sites can be grouped into three broad archetypes:

- **Building-based site:** Uses an existing rooftop or elevated structure. It may offer strong access but require structural, safety, power, and circulation upgrades.
- **Integrated hub:** Provides multiple pads or stands together with charging, processing, maintenance, and passenger or cargo facilities.
- **Ground-based site:** Uses available land and may be easier to expand, but can create longer access time to major customer locations.

These are classroom archetypes rather than an official Hong Kong classification. Actual design depends on aircraft, operations, safety requirements, and local approval.

Site Selection: A Multi-Criteria Decision

A suitable site must balance several considerations:

- **Safety and obstacle environment:** approach and departure paths, separation, downwash or outwash, emergency response, and surrounding uses.
- **Customer access:** total door-to-door time, transfers, elevators, walking, and accessibility.
- **Utilities and operations:** charging, communications, maintenance, storage, staffing, and recovery capability.
- **Capacity and expandability:** pads, parking or stands, turnaround time, operating hours, and future growth.
- **Cost and approval:** construction, adaptation, lease, environmental effects, community acceptance, and regulatory requirements.

Applied Tool: Comparing Sites

A Simple Annual Site-Cost Framework

$$\text{Generalized Site Cost} = \text{Annualized Infrastructure Cost} + \text{Annual OPEX} \\ + \text{Monetized Access Burden} + \text{Expected Disruption}$$

- Infrastructure cost includes construction and major site adaptation, net of residual value where relevant.
- OPEX includes staffing, utilities, maintenance, insurance, communications, and recurring site costs.
- Access burden converts additional passenger or cargo access time into a comparable value.
- Disruption cost may reflect weather, maintenance, congestion, or other reliability risks.

Lower generalized cost is not automatically better if the site cannot meet safety, capacity, or service requirements.

Back-of-the-Envelope: Comparing Two Site Archetypes

Assumptions: 100 passenger-trips per day and value of access time = HK\$15 per minute.

Building-Based Site

- Daily infrastructure allocation: HK\$5,000
- Daily OPEX: HK\$8,000
- Extra access time: 8 minutes
- Access burden:
 $8 \times 15 \times 100 = \text{HK\$12,000}$
- **Generalized site cost:**
HK\$25,000/day

Ground-Based Site

- Daily infrastructure allocation: HK\$1,000
- Daily OPEX: HK\$2,000
- Extra access time: 3 minutes
- Access burden:
 $3 \times 15 \times 100 = \text{HK\$4,500}$
- **Generalized site cost:**
HK\$7,500/day

Takeaway: Under these assumptions, the ground site has lower generalized cost. Actual choice also depends on demand, safety, capacity, network value, and feasible access.

Illustrative assumptions for classroom analysis; not current site quotations.

Concept Primer: Throughput and Reliability

Throughput is the number of aircraft movements, passengers, or missions that a facility can process during a defined period.

- It depends on pads, stands, turnaround, charging, passenger processing, staffing, airspace, and operating procedures.

Reliability is the facility's ability to provide the planned capacity when required.

- Downtime reduces usable capacity and may create delays, diversions, refunds, or lost demand.

Managerial lesson: Theoretical maximum throughput is less useful than reliable, serviceable capacity under realistic conditions.

Applied Tool: Breakeven Capacity Use

A Simple Facility Formula

$$\text{Breakeven Capacity Use} = \frac{\text{Fixed Daily Facility Cost}}{\text{Reliable Daily Capacity} \times \text{Contribution per Movement}}$$

- Contribution per movement is the facility fee minus variable cost associated with that movement.
- Reliable capacity should reflect operating hours, turnaround, maintenance, weather, priority access, and other constraints.
- The result is the share of reliable capacity that must be sold to cover fixed facility cost.
- It is not equivalent to the percentage of hours in the day during which the facility must operate.

Back-of-the-Envelope: Breakeven Capacity Use

Assumptions: Reliable capacity = 100 movements/day; facility fee = HK\$200/movement; variable facility cost omitted.

Higher-Cost Site

- Fixed cost: HK\$15,000/day
- Maximum fee revenue: HK\$20,000/day
- Breakeven use: 15,000/20,000
- **75% of reliable capacity**

Lower-Cost Site

- Fixed cost: HK\$4,000/day
- Maximum fee revenue: HK\$20,000/day
- Breakeven use: 4,000/20,000
- **20% of reliable capacity**

Takeaway: A higher-cost site requires more activity, a higher contribution per movement, or complementary revenue.

Illustrative assumptions; excludes variable cost, time-of-day demand, and service mix.

Concept Primer: Staged Ramp-Up

New infrastructure rarely reaches mature demand immediately.

- Early utilization may be limited by customer adoption, aircraft availability, approvals, operating experience, and network coverage.
- A staged plan links investment to evidence and milestones rather than building the full facility at once.
- Possible stages include a temporary or modular site, limited operating hours, selected users, and later expansion.
- The financial plan must include working capital and liquidity for early operating losses and contingencies.

Managerial question: Which investment is reversible or expandable if demand develops differently from the forecast?

Back-of-the-Envelope: Funding the Ramp-Up

Assumptions: Initial infrastructure investment = HK\$10 million; annual fixed operating cost = HK\$2 million. Other operating costs are omitted.

Mature-Demand Case

- Facility use: 80%
- Annual revenue: HK\$4 million
- Fixed operating cost: HK\$2 million
- **Operating surplus before other costs: HK\$2 million**

Year 1 Ramp-Up Case

- Facility use: 20%
- Annual revenue: HK\$1 million
- Fixed operating cost: HK\$2 million
- **Operating shortfall before other costs: HK\$1 million**

Takeaway: The financing plan must cover the expected ramp-up shortfall plus a contingency buffer.

Illustrative assumptions; excludes variable cost, financing, depreciation, tax, and working-capital timing.

Applied Tool: A Reliability Scenario

A Simple Annual Return Screen

$$\text{Simple Return} = \frac{\text{Adjusted Revenue} - \text{Adjusted Operating Cost}}{\text{Initial Investment}}$$

- Adjusted revenue should reflect serviceable capacity, demand, cancellations, and pricing under the scenario.
- Some costs fall during downtime, while fixed cost, recovery cost, refunds, and service credits may remain or increase.
- Weather should be modelled from route- and aircraft-specific operating assumptions rather than a universal annual percentage.
- This simple return is not a full ROI or NPV analysis because it ignores timing, financing, tax, and residual value.

Back-of-the-Envelope: A Downtime Scenario

Assumptions: Initial investment = HK\$10 million; annual revenue without disruption = HK\$4 million; annual cost = HK\$2.5 million.

No-Downtime Scenario

- Operating surplus: HK\$1.5 million
- Simple return: 1.5/10
- **15%**

20% Revenue Reduction

- Adjusted revenue: HK\$3.2 million
- Assumed cost: HK\$2.5 million
- Operating surplus: HK\$0.7 million
- **Simple return: 7%**

Takeaway: The assumed disruption materially affects returns. A better model would separate capacity loss, demand loss, variable-cost savings, and recovery cost.

The 20% reduction is an illustrative stress scenario, not a claim about Hong Kong annual weather downtime.

Session 2

CAPEX, OPEX and the Maintenance Money-Pit

Concept Primer: Infrastructure Investment

Vertiport investment may include:

- site acquisition or adaptation and structural works;
- touchdown, lift-off, safety, parking, and passenger or cargo areas;
- charging, energy storage, communications, lighting, security, and fire protection;
- access, circulation, accessibility, and integration with ground transport; and
- design, engineering, testing, approval, and contingency allowances.

Requirements depend on aircraft characteristics, operating concept, site conditions, and the applicable regulatory framework.

Concept Primer: Infrastructure OPEX

Recurring facility cost may include:

- staffing, security, emergency response, and passenger or cargo processing;
- electricity, communications, software, monitoring, and data services;
- inspection, preventive maintenance, repairs, cleaning, and replacement reserves;
- lease payments, insurance, professional services, and recurring compliance activities; and
- disruption recovery, customer support, and service-level obligations.

Managerial lesson: Do not assume electricity is the dominant OPEX item. The cost mix depends on the site and operating model.

Applied Tool: Daily Infrastructure Cost

A Simple Economic-Cost Approximation

$$\text{Daily Infrastructure Cost} = \frac{\text{CAPEX} - \text{Expected Residual Value}}{\text{Useful Operating Days}} + \text{Daily OPEX}$$

- The formula allocates net investment cost over the expected operating life.
- It is useful for unit-cost screening but is not identical to daily cash flow, accounting depreciation, mortgage payment, ROI, or NPV.
- A complete investment model should also consider financing, replacement cycles, tax, residual value, and the timing of cash flows.

Back-of-the-Envelope: Daily Infrastructure Cost

Assumption: Ten-year life and no residual value.

Higher-Investment Site

- CAPEX: HK\$18.25 million
- Daily allocation: HK\$5,000
- Daily OPEX: HK\$10,000
- **Daily economic cost:**
HK\$15,000

Lower-Investment Site

- CAPEX: HK\$3.65 million
- Daily allocation: HK\$1,000
- Daily OPEX: HK\$3,000
- **Daily economic cost: HK\$4,000**

Takeaway: A higher-cost site must create sufficient additional demand, price, reliability, network value, or complementary revenue to justify the difference.

Illustrative assumptions; excludes financing, tax, residual value, and replacement cycles.

Session 3

Slot Allocation, EMS Priority & GBA Scale

Concept Primer: Allocating Scarce Slots

When requests exceed safe and reliable capacity, managers need a transparent allocation rule.

- **Administrative allocation:** fixed schedules, permits, historic rights, rotation, or first-come-first-served.
- **Market-based allocation:** posted peak pricing or auctions where appropriate.
- **Priority rules:** emergency, public-service, safety, or other protected operations may receive priority under predefined procedures.
- **Operational controls:** reservations, buffers, delay management, diversions, and cancellation rules.

Managerial objective: Balance safety, efficiency, predictability, access, competition, and public-service obligations.

Applied Tool: A Peak-Pricing Rule

An Illustrative Posted-Price Rule

$$\text{Slot Fee} = \text{Base Fee} + \text{Congestion Charge} \left(\frac{\text{Requests}}{\text{Available Slots}} \right)$$

- The exact pricing function is a managerial design choice, not a standard regulatory formula.
- Peak charges may shift flexible demand, but only if operators can change timing, route, or facility.
- Pricing must be coordinated with safety limits, priority rules, competition policy, transparency, and access obligations.
- Revenue maximization should not be the only objective of scarce infrastructure allocation.

Back-of-the-Envelope: An Illustrative Peak Fee

Assumptions: Base fee = HK\$100 and an illustrative congestion parameter = HK\$200.

Two Requests, Two Slots

- Demand-capacity ratio: $2/2 = 1$
- Fee: $100 + 1(200)$
- **HK\$300**

Eight Requests, Two Slots

- Demand-capacity ratio: $8/2 = 4$
- Fee: $100 + 4(200)$
- **HK\$900**

Takeaway: The higher fee may shift flexible trips, but customer response must be estimated. Price alone does not identify which missions create the greatest social value.

Illustrative pricing rule and assumptions; not a recommended tariff.

Hong Kong and the GBA: Infrastructure Possibilities

Dimension	Hong Kong possibility	Wider GBA possibility
Site environment	Dense and constrained	More varied
Customer access	Strong value in central locations	Varies by city and country
Infrastructure role	Integration and specialized services	Larger network and production
Likely design	Hub, distributed, or hybrid	Hub, distributed, or hybrid
Key question	Value created per scarce site	Network coverage and capacity

Network design should be based on route-specific demand, site feasibility, aircraft, regulation, and access rather than a predetermined Hong Kong or GBA model.

Concept Primer: Network Design

Should the network concentrate activity in a few hubs or distribute activity across many sites?

- **Concentrated hub network:** May support shared facilities, high throughput, connections, and specialized services, but can increase access time and vulnerability to disruption.
- **Distributed network:** May improve customer access and resilience, but can duplicate fixed cost and reduce activity at each site.
- **Hybrid network:** Combines major hubs with smaller feeder or specialized sites.

Managerial lesson: There is no universal “Hong Kong strategy” or “GBA strategy.” The best configuration depends on demand, geography, capacity, and service purpose.

Applied Tool: Comparing Network Designs

Generalized Cost per Served Trip

$$\text{Generalized Cost per Trip} = \frac{\text{Network Fixed Cost} + \text{Network OPEX} + \text{Disrupt}}{\text{Served Trips}}$$

- Concentration may lower facility cost per trip but increase passenger or cargo access burden.
- Distribution may improve access and resilience while increasing duplicated cost.
- Served trips, not unconstrained forecast demand, should be used in the denominator.
- Compare service quality, reliability, expansion options, and network connections as well as cost.

Back-of-the-Envelope: Concentrated vs Distributed Network

Assumptions: 4,000 served passenger-trips/day and value of access time = HK\$15/minute.

Concentrated Network

- Daily network cost: HK\$15,000
- Cost per trip: HK\$3.75
- Access burden:
 $8 \times 15 = \text{HK\$120/trip}$
- **Generalized cost:**
HK\$123.75/trip

Distributed Network

- Daily network cost: HK\$45,000
- Cost per trip: HK\$11.25
- Access burden:
 $2 \times 15 = \text{HK\$30/trip}$
- **Generalized cost: HK\$41.25/trip**

Takeaway: The distributed design wins under these assumed volumes and access times. The result may reverse if demand, reliability, feasible sites, or network cost changes.

Illustrative assumptions; 4,000 refers to passenger-trips, not flights.

Activity: Stress-Testing a Two-Pad Facility

- **Format:** Groups of 4–5
- **Base task:** Estimate reliable daily capacity, contribution per movement, breakeven capacity use, and simple annual return.
- **Scenario 1:** Reduce serviceable capacity to reflect weather or maintenance downtime.
- **Scenario 2:** Reserve five daily slots for priority public-service use.
- **Decision:** Recommend a fee, availability payment, subsidy, or cost reduction that keeps the project viable without violating capacity or priority rules.
- **Timing:** 15 minutes plus 5-minute discussion.

Key Takeaways

- Infrastructure strategy balances safety, customer access, utilities, reliable capacity, cost, expandability, and community impact.
- Breakeven should use reliable capacity and contribution per movement, not theoretical maximum activity or gross fees alone.
- Staged investment and liquidity planning reduce the risk of building ahead of demand.
- Weather and maintenance should be modelled through explicit capacity, demand, cost, and recovery assumptions.
- Scarce slots require transparent allocation and priority rules; peak pricing is only one possible mechanism.
- Hub, distributed, and hybrid networks each involve trade-offs between facility cost, access, resilience, and connectivity.

Infrastructure Viability Model and Memo (maximum 1,200 words plus spreadsheet)

- **Due:** Beginning of Week 6
- **Task:** Evaluate a building-based, ground-based, or integrated-hub archetype for your selected use case.
- **Requirements:**
 1. Explain the site-selection criteria and define reliable daily capacity.
 2. Separate infrastructure investment, fixed OPEX, variable facility cost, and access burden.
 3. Calculate breakeven capacity use and a simple annual return.
 4. Model a staged ramp-up and one weather or maintenance scenario.
 5. Propose a slot-allocation rule that protects priority services.
 6. Compare the selected design with one alternative and recommend whether to build, stage, relocate, or reject the project.

Next Week Preview:

Public-Private Partnerships: Who pays for the chargers?